SAP Business Network for Logistics – REST API Integration series: Post #2



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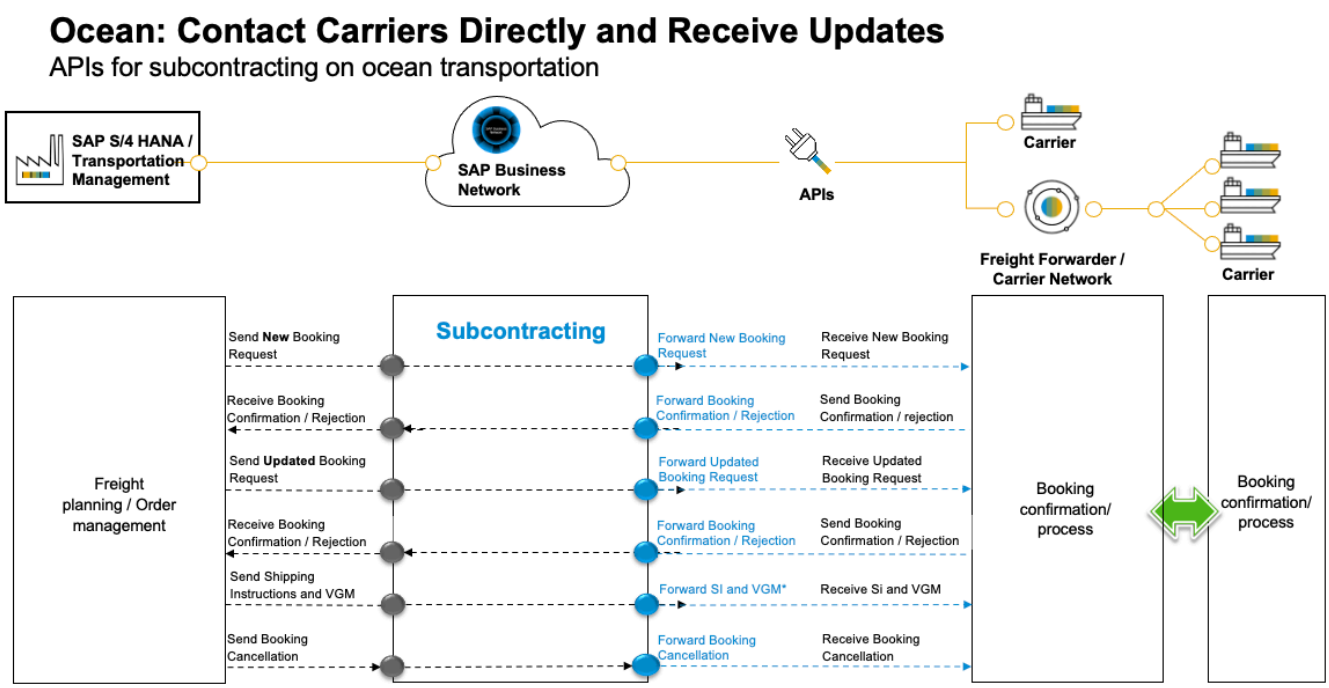
Breaking down the Freight Booking API (Ocean) for Carrier Integration (2/2)

Glossary:

Acronym	Definition
BN4L	Business Network for Logistics
LBN	Logistics Business Network (old name for BN4L)
TM	Transportation Management
API	Application Programming Interface
MIL	Manage Integration Logs
VGM	Verified Gross Mass

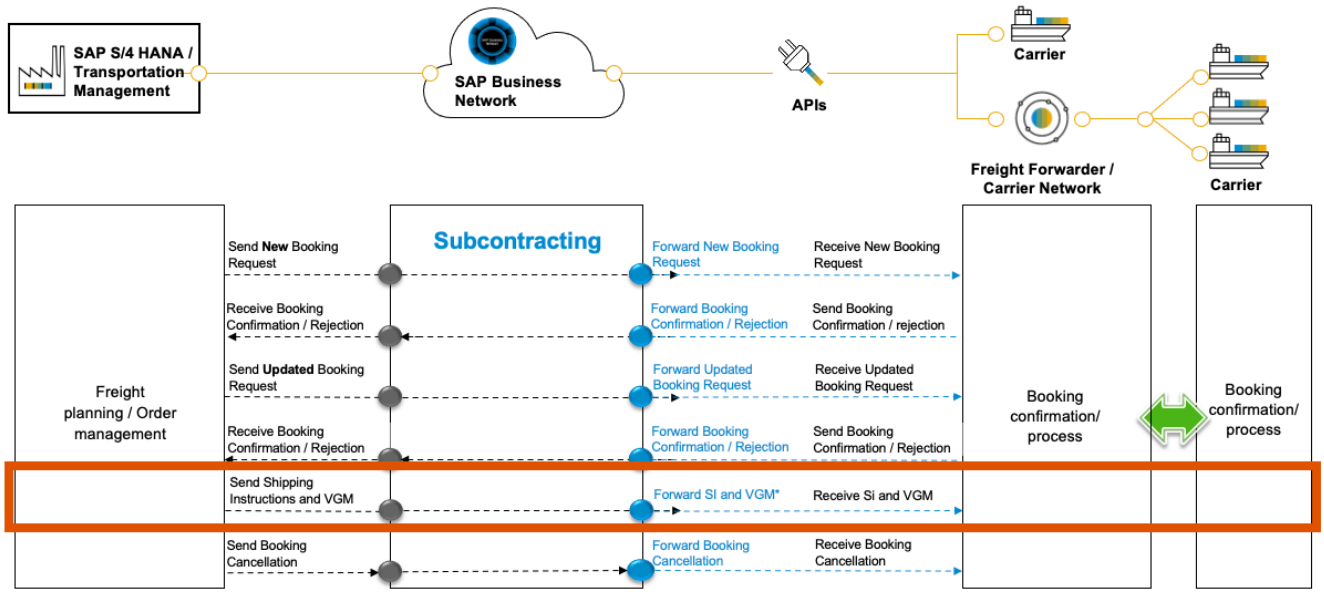
The SAP Business Network Freight Collaboration solution connects shippers and their logistics partners on a secure network for optimized collaboration and insights, integrated tightly with core business processes for efficiency and resilience. More details can be found here.

The Freight booking document is a primary transportation document for Ocean and Air. The processes supported in SAP BN4L for Ocean scenario are explained here. In this post, the focus is on usage of booking API and the minimum data that is required for a successful end-end integration.



Booking Subcontracting process

Use Case 2: Shipping Instructions and VGM



Shipping Instructions and Booking Cancellation

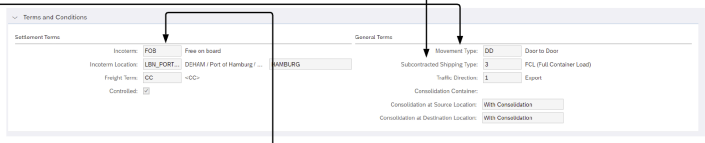
Waybill notification is sent by the Shipper through their TM system and the Carriers get the Shipping Instructions and VGM through the FreightBookingWaybillNotificaiton API. This is a

POST call that creates and also sends SI & VGM updates. An example of a shipping instruction.

Some of the key fields and the breakdown of the exemplary payload is shown below:

Note: The data in the waybill notification is subject to the level of detail a shipper maintains and shares with the carriers.

```
object {7}
  documentRecipientParty {2}
  transportationOrderBookingWaybillNotification {38}
    shippingTypeCode : 3
    transportationStages [3]
    movementTypeCode : 1
    referenceCurrencyCode : value
    consigneeParty {4}
    grossWeightTotalQuantity {2}
      unit : KGM
      value : 2500
    carrierParty {4}
    actionCode : 02
    baseBusinessTransactionDocumentReference {4}
    insuranceDeclaredTotalAmount : null
    transportationTerms {5}
      lbnTransportationModeCode : null
      specialHandlingCodes [0]
      transportModeCode : 1
      transportServiceLevelCode : null
      additionalTransportServiceLevelCode : null
    deliveryTerms {1}
      incoTerms {2}
        incotermClassificationCode : FOB
        incotermTransferLocation {4}
          address {13}
          UNLOCODELocationText : null
          timeZone : CET
          id : LBN_PORT_HH
    ordinalNumber : 4
    additionalParty [4]
    dangerousGoods : true
    externalDocumentIdentifier : BNC5839392
    grossVolumeTotalQuantity {2}
      unit : MTQ
      value : 2500
    orderingParty {4}
    businessTransactionDocumentReferences [1]
      0 {3}
        transportationDocumentTypeCode : T67
        issueDateTime {3}
        businessTransactionDocumentReference {4}
          itemId : null
          itemTypeCode : null
          id : BNC5839392
          typeCode : null
    totalBookedCapacityVolumeQuantity : null
    totalBookedCapacityPiecesQuantity : null
    categoryCode : null
    deliveryTotalAmount : null
    otherChargesPaymentArrangementCode : C
    piecesTotalQuantity : null
    regulatedAgentParty : null
    shippersLoadAndCountTotalQuantity : null
    textCollection [5]
      0 {4}
        text : General Information
        type : DFLT
        languageCode : null
        creationDateTime {3}
          LocalDateTime : 2022-12-28T03:20:15
          DateTimeUTC : 2022-12-28T03:20:15Z
          LocalTimezone : UTC
      1 {4}
        text : SEAWAY BL
        type : BOLTY
        languageCode : null
        creationDateTime {3}
          LocalDateTime : 2022-12-28T04:29:47
          DateTimeUTC : 2022-12-28T04:29:47Z
          LocalTimezone : UTC
      2 {4}
        text : 3
        type : BLCNO
        languageCode : null
        creationDateTime {3}
          LocalDateTime : 2022-12-28T06:36:28
          DateTimeUTC : 2022-12-28T06:36:28Z
          LocalTimezone : UTC
      3 {4}
        text : YES
        type : RATED
        languageCode : null
        creationDateTime {3}
          LocalDateTime : 2022-12-28T06:36:28
          DateTimeUTC : 2022-12-28T06:36:28Z
          LocalTimezone : UTC
      4 {4}
        text : Information to be printed in BOL
        type : BOLTX
        languageCode : null
        creationDateTime {3}
          LocalDateTime : 2023-02-01T06:45:21
          DateTimeUTC : 2023-02-01T06:45:21Z
          LocalTimezone : UTC
    freightBookingId : 6300002559
    totalBookedCapacityWeightQuantity : null
    weightAndValuationChargesPaymentArrangementCode : C
    customsDeclaredTotalAmount : null
    shipperParty {4}
    houseWaybillReference [0]
      (empty array)
    contractConditionCode : D2D
    items [1]
      0 {44}
        buyersConsolContainerIndicator : false
        acceptableTemperatureMeasureInterval : null
        dangerousGoodsInfo [0]
          (empty array)
        destinationCountryCode : null
        seal [1]
          0 {2}
            issuingPartyRoleCode : SG
            id : 67893058
        description : null
        lengthMeasure : null
        serviceInfo : null
        widthMeasure : null
        exportProcessCustomsAuthorityID : null
        heightMeasure : null
        transportationGoodsIdentification [0]
          (empty array)
        parentItemId : null
        consigneeParty : null
        originBusinessTransactionDocumentInformation : null
        originCountryCode : null
        shippersLoadAndCountTransportationQuantity : null
        shipToLocation {4}
          address {13}
          UNLOCODELocationText : ESALB
          timeZone : EST
          id : LBN_NUTS_CL
        deliveryNetAmount : null
        baseBusinessTransactionDocumentReference {4}
          itemId : 1000010
          itemTypeCode : null
          id : 6300002559
          typeCode : 1122
```

- This object holds incoterms (International Commercial terms), useful in international shipping

- Planned quantities

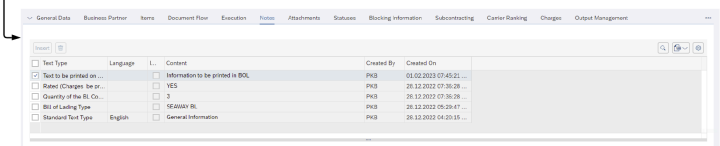
- References to Business docs

Text Type

You can introduce text types in your SAP TM or TM in SAP S/4HANA system to manage various settings.

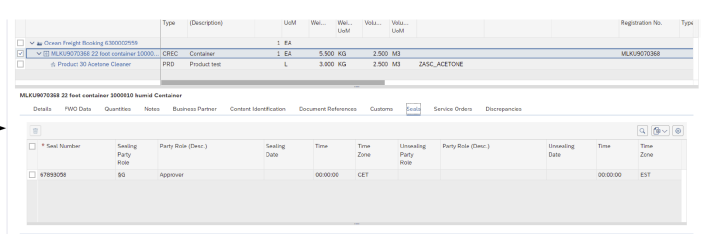
The following are the list of text types that you can use:

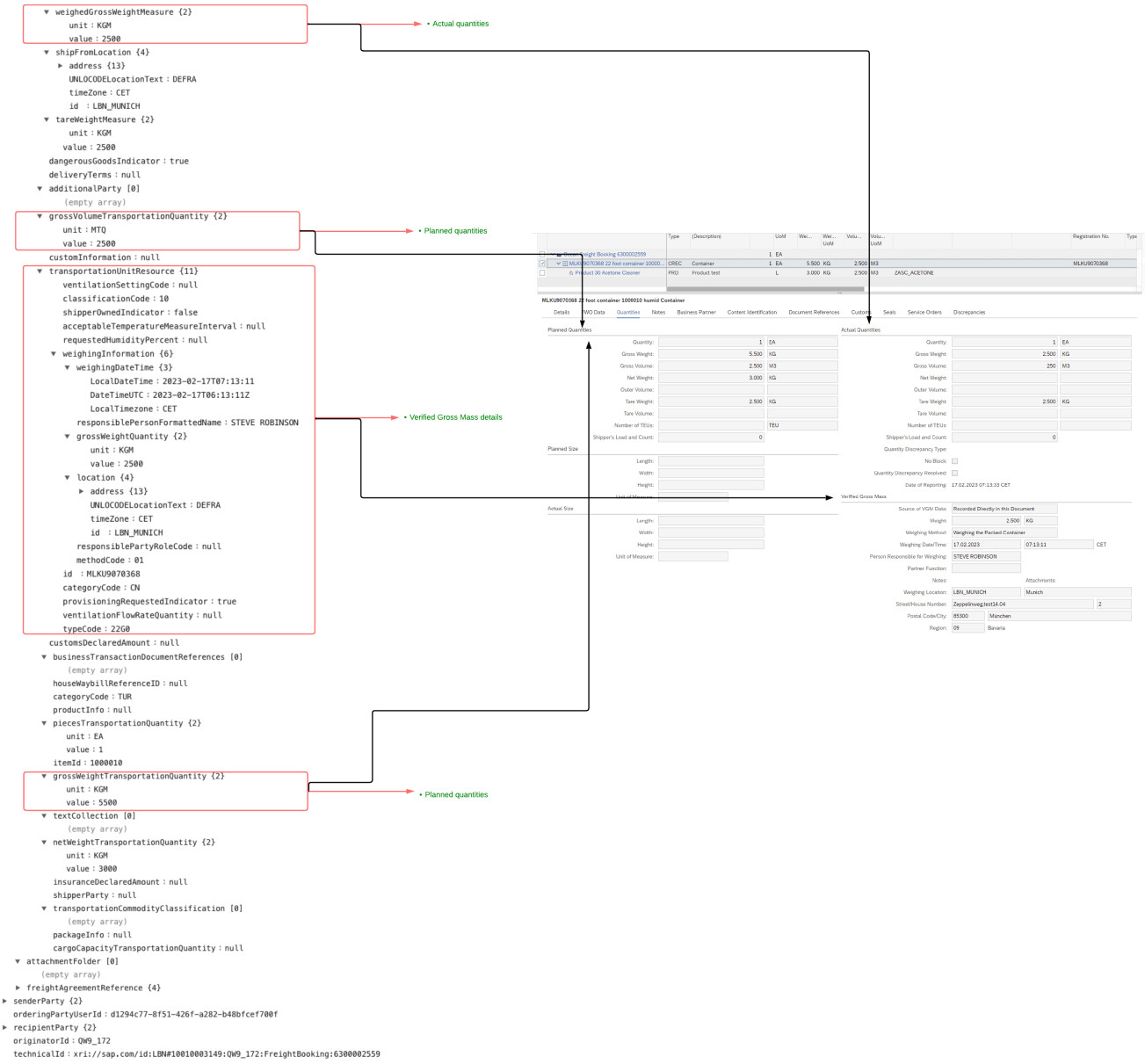
Text Type	Description	Purpose	Checkboxes to be Selected
E0103	Carrier's Note in Subcontr/Tendering	To allow the carrier to submit a note with the confirmation of a freight booking.	**InternalLang, Text**
TERML	Empty container Pick-Up Terminal Location	Shipper to share the terminal location from where empty container to be picked up.	**Lang, Text**
BOLTY	Original Bill of Lading vs Seaway (Copy) Bill of Lading	Indicates if the bill of lading is original or not. Two recommended values are **ORGINAL BL** or **SEAWAY BL**.	**Lang, Text**
BLCNO	Quantity of Bill of Lading Copy / Seaway Quantity	Indicates the quantity of bill of lading copy or the quantity of seaway.	**Lang, Text**
BLONO	Quantity of Original Bill of Lading	Indicates the quantity of original bill of lading.	**Lang, Text**
BOLTX	Free Text to be Printed on Bill of Lading	Indicates any free text to be printed on bill of lading.	**Lang, Text**



- Includes details about the dangerous goods provided the DG master data is maintained.

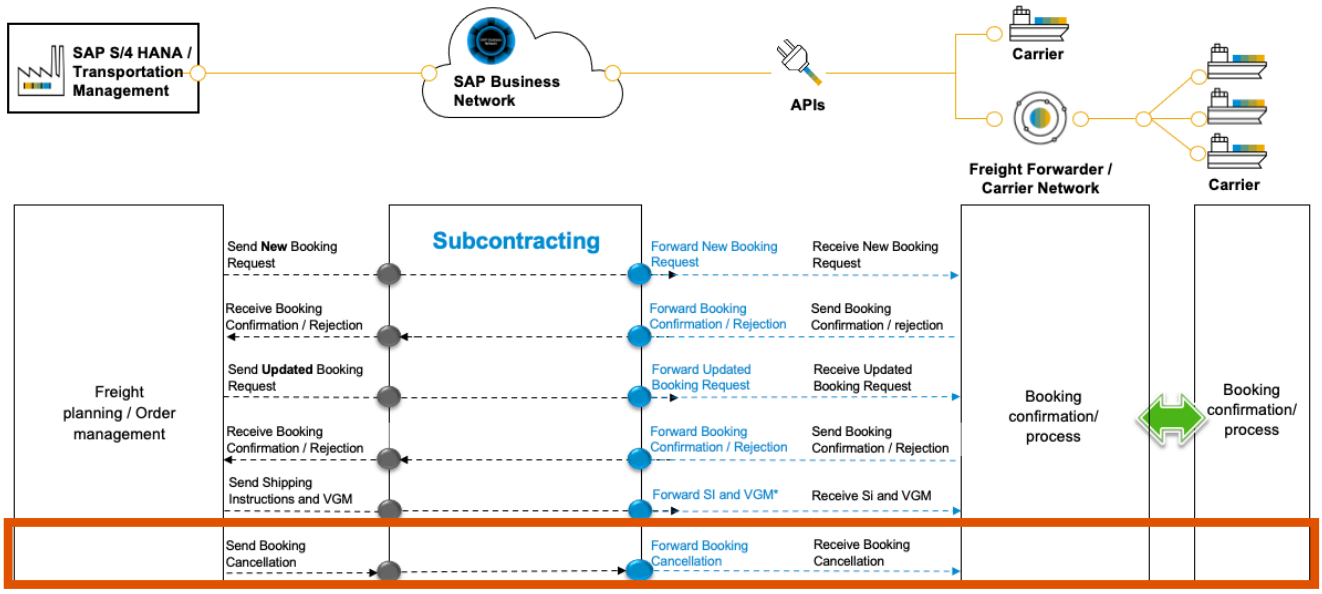
- seal details with issuing party.





Note: If you are not receiving dangerous goods information, see if you have implemented this note, which could be filtering out non-main items.

Use Case 3: Booking cancellation



Booking cancellation

Freight Bookings can be canceled by the Shipper and this Business event will be triggered by BN4L and delegated to Carriers and Network Partners via the consumer API for Booking Cancellation.

The example payload looks like below:

```
{
    "documentRecipientParty": {
        "lbnId": "10010001721",
        "standardIds": [{
            "schemeAgencyId": "TM",
            "schemeId": "",
            "value": "0001010581"
        }, {
            "schemeAgencyId": "182",
            "schemeId": "",
            "value": "LBNT"
        }, {
            "schemeAgencyId": "182",
            "schemeId": "",
            "value": "SCSF"
        }]
    },
    "transportationOrderBooking": {
        "ordinalNumber": 3,
        "externalDocumentIdentifier": "",
        "freightBookingId": "6300003001",
        "baseBusinessTransactionDocumentReference": {
            "id": "6300003001",
            "version": "0001010581-0003",
            "typeCode": "1122"
        }
    },
    "senderParty": {
        "lbnId": "10010001410",
        "standardIds": [{
            "schemeAgencyId": "TM",
            "schemeId": "",
            "value": "LBNFBSHP03"
        }, {
            "schemeAgencyId": "009",
            "schemeId": "",
            "value": "GLN00001"
        }]
    },
    "recipientParty": {
        "lbnId": "10010001721",
        "standardIds": null
    },
```

```
    "originatorId": "QW9_172",
    "technicalId": "xri://sap.com/id:LBN#10010001410:QW9_172:Freight
}
```

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